

ARCHERIDE_Deployment_and_Operations _Guide

Deployment & Operations Overview

ARCHERIDE is deployed as a production-grade, cloud-ready platform engineered for operational stability, scalability, and controlled public exposure.

Deployment Environment

- **Operating System:** Ubuntu 24.04 LTS
- **Hosting Model:** Private, secure cloud infrastructure
- **Capacity Profile:**
 - 4 vCPU
 - 16 GB RAM
 - 500 GB SSD storageOptimized to support real-time dashboards and analytical workloads.

Application Runtime

- **Backend Runtime:** Stateless Node.js API services
- **Frontend Delivery:** React application compiled into production-ready static assets
- **Proxy Layer:** Nginx serves static assets and proxies API traffic to backend services

Environment Configuration

- **Configuration Management:** Sensitive values (DB credentials, JWT secrets) are managed exclusively through environment variables
- **Environment Separation:** Development and production runtimes are isolated, with independent build and deployment pipelines

Availability & Stability

- **Process Management:** PM2 manages backend processes with automatic restarts on failure
- **Startup Persistence:** Applications are configured to restart automatically following system reboots
- **Health Monitoring:** A public `/api/health` endpoint supports uptime verification without authentication

Backup & Recovery

- **Snapshot Strategy:** Daily infrastructure snapshots of primary volumes
- **Database Backups:** Periodic database dumps for portability and recovery
- **Recovery Objectives:**
 - **RPO:** 24 hours
 - **RTO:** Less than 4 hours

Monitoring & Operational Integrity

- **Error Logging:** Server-side logs capture runtime and application errors for review
- **Health Checks:** Automated probes ensure synchronization between the reverse proxy and backend services

Change Management

- **Hotfix Deployment:** Targeted rebuilds of affected layers followed by controlled, zero-downtime restarts
- **Stability Assurance:** Every production update undergoes build validation prior to deployment

Public Exposure Readiness

- **Domain:** `ride.arche.global`
- **DNS Management:** Controlled DNS routing to production IP
- **Access Posture:** Platform is publicly accessible, with all functionality gated by RBAC and Business Manager authorization